Multi-Task Uplift Modeling & Incremental Incentive Optimization for Customer Conversion

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Abstract

Making users perform an action is a never-ending industrial problem and every industry invests a lot to increase user interaction with their products or services and also retain them. One of the widely used techniques for growth or retention is offering users incentives or promotions which can increase the likelihood of user conversion. In the machine learning space, there are many baseline conversion models that can predict the chances of conversion for a given incentive using user features but they ignore to understand whether the incentive actually changes user behavior. This leads to the wasted resources by offering incentives to users who would convert anyway. We propoase a project to build a machine learning model that reformulates customer-conversion targeting as an Individual Treatment Effect (ITE) estimation problem by using a multi tower neural network and a cost optimization function. We will evaluate the model using the open source data like Starbucks rewards.

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1 Introduction

1.1 Background and Motivation

Survival of every industry in the current competitive market is decided by the user and their level of engagement. Industries, on the other hand, alongside their development, they invest a lot in growing and retaining users. Modern growth strategies depend heavily on promotions and incentives to keep users engaged and in addition the companies are moving towards using the machine-learning models to predict the conversion probability such as $P(Y = 1 \mid X)$ based on user features. However, the trouble is that the outcome of the predictions does not reflect the actual impact. To elaborate, the incentives are allocated solely on the model-predicted conversion which can lead to two major issues. Unwanted incentives are given to the organic users with high intent to remain engaged regardless of the incentive and low-intent users getting the incentives.

These issues are prevalent because the models predicting the likelihood of the conversion do not consider the influence of incentives in users' behavior which can lead to budget misallocation. The concept of Individual Treatment Effect (ITE) measures the difference between the users behavior or the outcome with and without incentives. Therefore, our goal is to build a neural network to understand the individual treatment effect and can identify users who are unlikely to convert when incentives are not provided and user who are likely to convert when the incentives are given. As an outcome, this model can help industries to reduce the incentive misallocation and gives industries the ability to target the right users with more effective use of the incentive budget.

1.2 Causal Foundations and Uplift

To solve the problem of incentive misallocation, prediction strategies must think beyond standard outcome prediction and adapt to a causal framework, which focuses on predicting whether a user converts specifically because an incentive was provided [7,9]. The primary focus of standard models is estimating conditional conversion probabilities given user features, $P(Y = 1 \mid X)$. But when incentives are added to the modeling, we are not just observing baseline habits, but also how the introduction of incentives directly alters user choice.

To model this, we compare user outcomes when an incentive is offered ($T = 1$) with outcomes when nothing is offered ($T = 0$). That is, each user has two possible outcomes: $Y_i(1)$ (with

incentive) and $Y_i(0)$ (without incentive). The Individual Treatment Effect (τ_i) is the difference between these two outcomes, representing the direct influence of the incentive on the outcome.

$$\tau_i = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i] \tag{1}$$

where $\mathbb{E}[\cdot \mid X_i]$ denotes the expected value conditioned on the user's feature vector X_i.

In multiple incentives scenario or fixed and distinct incentive choices $T_k \in \{T_0, T_1, \ldots, T_K\}$, the increase in the conversion probability caused mainly by giving user specific incentive k. Which is denoted as:

$$\text{Uplift}_k(X_i) = P(Y_i = 1 \mid X_i, T = k) - P(Y_i = 1 \mid X_i, T = 0) \tag{2}$$

The major challenge of causal inference is that we can only observe one outcome for each user—that is, either $Y_i(1)$ or $Y_i(0)$—but cannot know the outcome for both at the same time [8]. As we can only test one choice per user, we are missing the other half. We use uplift modeling [12], which addresses this counterfactual problem by evaluating a large user population where an incentive is offered to a treatment group and nothing to the control group. The measured difference in conversion rates between treatment and control groups estimates the net causal effect of the incentive.

For these targeting problems, the user population can be categorized into four groups based on their outcomes:

Table 1: Behavioral Segmentation in Causal Uplift Modeling

Segment	$Y_i(0)$	$Y_i(1)$	**Business & Operational Impact**
Persuadables	0	1	**Primary Target.** Converts only if given an incentive; maximizes ROI.
Sure Things	1	1	**Do Not Target.** High intent; Saves the budget.
Lost Causes	0	0	**Do Not Target.** Disengaged; offering incentive yields zero return.
Sleeping Dogs	1	0	**Avoid.** Negative reaction; Incentives created user drop the intent.

Using the above user segmentation, standard machine learning models target both "Persuadables" and "Sure Things" because users in these groups exhibit high conversion probabilities, leading to budget misallocation when incentives are given to "Sure Things". On the other hand, the uplift model targets only the "Persuadables" user segment, where conversion is higher specifically when an incentive is provided.

Standard machine learning models are built to predict a single outcome for given features, but they do not support calculating causal differences. That is, with respect to the problem in

this proposal, traditional models cannot isolate the true conversion impact when an incentive is provided. To solve this, we can use meta-learners [4]. Meta-learners use standard machine learning models under the hood by arranging them in clever ways that can ultimately calculate the causal difference. Below are the standard meta-learner approaches:

- **S-Learner (Single Model):** A single prediction model $f(X, T)$ uses the original user features (X) and the incentive flag (T) as another feature. The uplift ($\hat{\tau}(X)$) is estimated by taking the difference of outcomes from the model used to predict the conversion score with the incentive ($T = 1$) and without the incentive ($T = 0$). On the other hand, the incentive flag alongside the other features may be ignored if the original feature set is very large.

- **T-Learner (Two Models):** Two prediction models are used where one model $\mu_1(X)$ is trained on the treatment data of the user who are presented with the incentive and the other model $\mu_0(X)$ trained on the control data of the user without an incentive. Where the uplift is calculated as $\hat{\tau}(X) = \mu_1(X) - \mu_0(X)$. This model can reduce the treatment suppression but can suffer from the data imbalance between the arms.

- **X-Learner (Crossover Learner):** This fixes the data imbalance problem and works in two stage. First guessed the counterfactual outcome of every user and later train the model on the guessed difference [4].

These meta-learners are great as a baseline, but struggle with complex data with large feature set. As a mitigation, deep neural networks with shared features and separate treatment heads are needed.

1.3 Scope and Objectives

- **Primary Goal:** Design and evaluate a deep multi-tower neural network for causal uplift modeling across multiple incentives.

- **Core Objectives:**

  - Build a shared feature trunk with dedicated output head for each discrete incentives ($T \in \{T_0, \ldots, T_K\}$).

  - Compare performance against standard baselines model using uplift ranking curves.

- **In-Scope:** Offline dataset processing, model hyperparameter tuning, and comparative ranking performance analysis.
- **Out-of-Scope:** Live production serving microservices and active online A/B testing.

2 Problem Statement

2.1 Mathematical Formulation of Uplift

Let $X \in \mathbb{R}^d$ represent observed user features, $T \in \{0, 1, \ldots, K\}$ denote the assigned treatment arm($T = 0$ for the arm with the users without incentive; T_k for users with the incentive of the tier k), and $Y \in \{0, 1\}$ represent the binary outcome of the user conversion.

In reference to the Rubin-Neyman Potential Outcomes Framework [7], each user will have a potential outcomes $Y(T = k)$ irrespective of the incentive tier. The Individual Treatment Effect (ITE) for any incentive tier k relative to control is:

$$\tau_k(X) = \mathbb{E}[Y(T = k) \mid X] - \mathbb{E}[Y(T = 0) \mid X] \tag{3}$$

2.2 The Fundamental Problem of Causal Inference

Because individual counterfactual outcomes are unobserved [8], models rely on three structural assumptions. These condition will fill the gap between observed data and the causal inference. This will ensure the model estimates the real treatment influence instead of unknown selection bias.

- **Unconfoundedness:** No hidden bias. Incentive assignments should only depend on the user features (X) that are available to see and measure [10].

  $(Y(0), Y(1), \ldots, Y(K)) \perp T \mid X$

- **Positivity:** For every type of user profile with similar features, there must be a non-zero probability of receiving any incentive tier or no incentive.

  $0 < P(T = k \mid X) < 1, \quad \forall k \in \{0, \ldots, K\}$

- **SUTVA:** The Stable Unit Treatment Value Assumption requires that the incentive to one user should not impact the outcome of the other user [11].

2.3 Segmentation Quadrants

Based on the potential outcome combinations defined in Section 1.2 (Table 1), uplift modeling will only target the **Persuadables** segment ($Y(0) = 0, Y(1) = 1$) to maximize conversion and exclude the rest as they do not show any positive impact to the incentive.

3 Related Work

3.1 Traditional Response Modeling vs. Causal Modeling

The proposal states that the standard machine learning models are not suitable for calculating the causal difference and instead focus on the outcome probability $P(Y = 1 \mid X)$. The literature also demonstrates the same thing that maximizing prediction accuracy does not correlate with maximizing incremental revenue [5].

3.2 Causal Meta-Learners

As described in Section 1.2, standard meta-learners (S-, T-, and X-Learners) use standard machine learning models to estimate the Individual Treatment Effects [4]. These meta-learners are useful for smaller datasets and feature sets whereas for high-dimensional feature spaces these model will struggle and become inefficient when applied to multiple incentive tiers.

3.3 Deep Learning for Uplift Modeling

The proposal is to build a deep neural network to solve this incentive allocation problem. We want to use the modern deep learning models like TarNet and CFRNet to estimate the causal effect. These models use a shared base to learn balanced representation across all the treatment arms [13]. Additionally, these models come with the built in domain regularization and penalty functions to eliminate representation discrepancies and achieve fair causal comparison across all the treatment arms [14].

4 Proposed Solution

4.1 Multi-Tower PyTorch Deep Neural Network Architecture

We propose a Multi-Tower Deep Neural Network build for multiple incentive allocation.

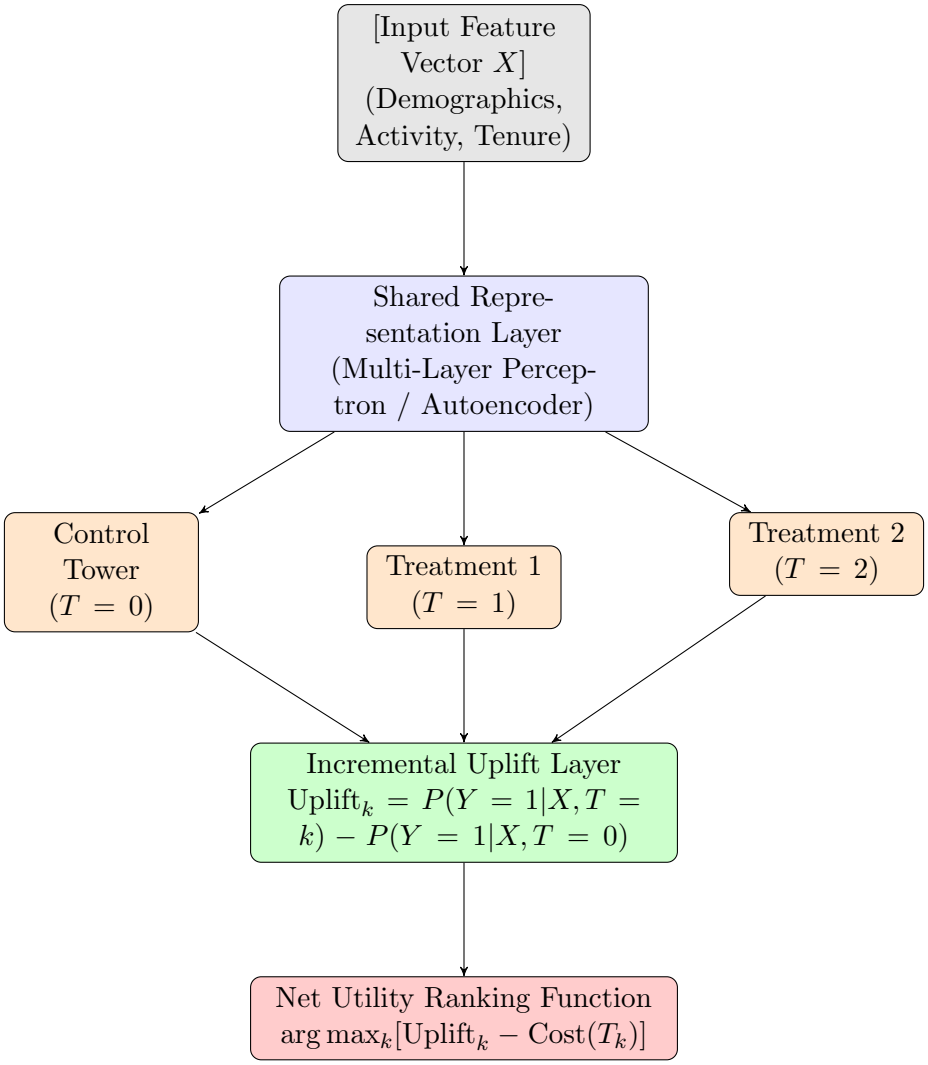


Figure 1: Multi-Tower Architecture for Multiple Incentive Optimization.

4.2 Net Utility Ranking Engine

The incentive with higher tier of benefits can have higher conversion compared to the other tiers. Ranking higher tier more can impact the incentive revenue. So instead of assigning incentive ranking based on maximum uplift, the model will include cost parameters $C(T_k)$ to each incentive tier k:

$$k^* = \arg \max_{k \in \{0,\ldots,K\}} (\hat{\tau}_k(X) \cdot V - C(T_k)) \tag{4}$$

where V represents monetary conversion value, and $C(T_k)$ is the margin cost of the incentive

5 Dataset and Preprocessing Pipeline

5.1 Dataset Overview

The evaluation will use an open-source data **Starbucks Rewards Offer Dataset** [1]. The dataset include the user behavior over a 30-day window on the mobile application. The data is distributed into three relational files:

- **Portfolio:** Incentive metadata including tier, minimum spend threshold, incentive value, validity duration, and distribution channels.
- **Profile:** Customer demographic features (X) including age, income level, gender, and membership tenure.
- **Transcript:** Timestamped event logs tracking incentive receipts, views, completions, and general transaction amounts.

5.2 Trajectory-Based Cohort Construction

The proposal is primary focused on building the causal uplift model to understand the influence on the conversion when the users presented with and without the incentives. On the other hand, all users in the dataset receive incentives and the explicit no incentive control group is unavailable. We will look at each user's event history to know whether the user have viewed any incentive before making the transaction and accordingly will separate them into true treatment group (users with incentive) and control group (users without incentives).

- **Treatment Cohort (T_k):** Users who received incentives k, *actively viewed it*, and completed the required spend within the valid window.
- **Control Cohort (T_0):** Users who received an incentive but *never viewed it* prior to transacting, isolating uninfluenced organic conversion $Y(0)$.

6 Proposed List of Experiments

6.1 Baseline Models

To evaluate the efficiency of the proposed Multi-Tower Deep Neural Network, we compare its performance against distinct baseline models:

- **Single-Target Standard Classifiers:** The baseline models like logistic regression, LightGBM, etc. are trained directly on the conversion outcome Y using the incentive treatment T_k as a standard input feature. The models focus on the simple outcome prediction without including the causal counterfactual reasoning.
- **Standard Causal Meta-Learners:** Using the established meta-learning algorithms implemented via `CausalML` [2] and `EconML` [3]. The techniques we want to use is covered in Section 1.2.

6.2 Ablation Studies

To isolate the contribution of individual architectural design choices, we propose the following systematic ablation experiments:

- **Shared Representation Initialization:** - We will evaluate if the autoencoder pretraining helps the model to understand the user behavior compared to the standard layers where we initialize the network with random weights and train the entire model at once followed by learning the features and uplift predictions together.
- **Propensity Weighting Loss Mechanics:** We will evaluate the Inverse Propensity Score (IPS) weighted loss to improve the model estimation. This will help the model to avoid targeting the organic user with the incentive. [16].

7 Experiment Setup Details

7.1 Software and Hardware Specifications

- **Software:** The deep neural architectures will be built using Python 3.10+ and PyTorch 2.x including Scikit-Learn and LightGBM for baseline modeling. The open-source CausalML [2] package for Causal benchmarking and EconML [3] package for meta-learning.
- **Hardware:** The model training and causal analysis will be executed locally on an Apple Mac mini (M4 Pro, 14-core CPU, 20-core GPU, 64 GB Unified Memory). Additionally, the experiment will use PyTorch Metal Performance Shaders (`mps`) for hardware acceleration.

8 Evaluation Metrics

This section covers the performance evaluation. To evaluate the performance without the counterfactual information, we will rank everyone from most to least persuadable $\hat{\tau}(X_i)$ grouped by percentages $k \in (0, 1]$. For top-k groups, we will count the total number of the instances and how many converted. Let $N_{(k)}$ be the instances in the top-k groups, where $Y_t(k)$ and $Y_c(k)$ denote the number of conversion with the incentive ($T = 1$) and without the incentive ($T = 0$). Similarly, $N_t(k)$ and $N_c(k)$ represent sample sizes within the top-k groups.

8.1 Area Under the Uplift Curve (AUUC)

The uplift gain $g(k)$ at each group k measures the conversion rate achieved by targeting the top-k group

$$g(k) = \left(\frac{Y_t(k)}{N_t(k)} - \frac{Y_c(k)}{N_c(k)} \right) \cdot N_{(k)} \tag{5}$$

The **Area Under the Uplift Curve (AUUC)** is calculated by integrating $g(k)$ across all groups $k \in (0, 1]$:

$$\text{AUUC} = \int_0^1 g(k)\, dk \approx \sum_{j=1}^{K} g\left(\frac{j}{K}\right) \Delta k \tag{6}$$

where K is the number of groups used to evaluation.

8.2 Qini Curve and Qini Score

AUUC primarily evaluates the conversion differences but cannot count the users converted. We will use **Qini curve** $q(k)$ to measure the count of user converted.

$$q(k) = Y_t(k) - Y_c(k) \cdot \frac{N_t(k)}{N_c(k)} \tag{7}$$

The term $\frac{N_t(k)}{N_c(k)}$ corrects the imbalance in the class or tier variations between the groups.

The **Qini Score (Q)** represents the performance gain using the area between the model's Qini curve and a baseline random targeting ($q_{\text{random}}(k)$):

$$Q = \int_0^1 \left(q(k) - q_{\text{random}}(k)\right) dk \tag{8}$$

where $q_{\text{random}}(k) = k \cdot \left(Y_t(1) - Y_c(1) \cdot \frac{N_t(1)}{N_c(1)}\right)$ is the expected number of incremental conversions we get if we randomly target a group k of the total instances.

9 Detailed Project Execution Timeline

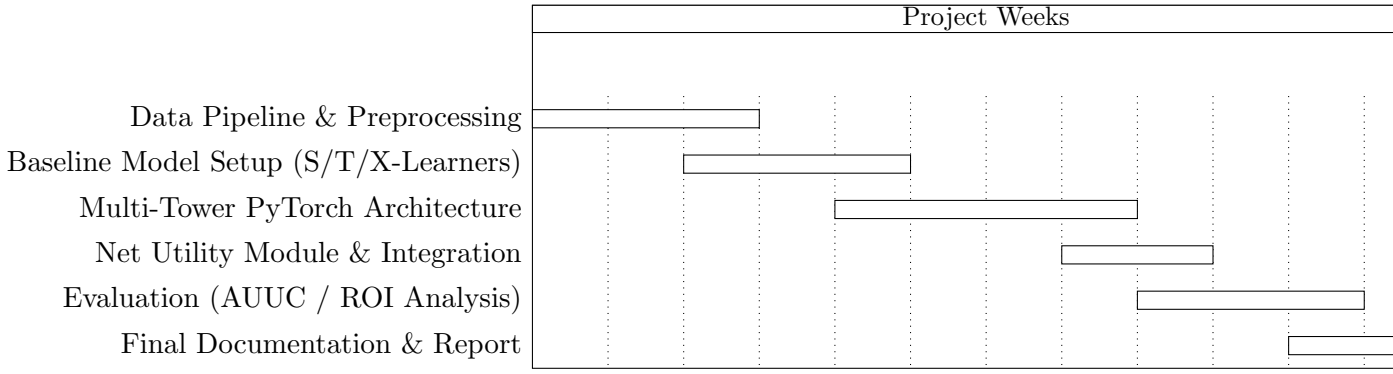


Figure 2: Gantt Chart of Proposed Project Timeline.

10 References

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